TOPPERS 24/7 OFFICIAL NOTES

Class 12 Biology

Comprehensive Priority Notes

All Must-Know Topics · TikZ Diagrams · Key Points

CBSE Board Exam 2025–26

very high | high | medium | low

Leadership: Lucky (Founder) | Tarun (Owner) | Krish (Director)
Mentors: Abhishek | Davneet | Aakarshit (Manager)
Community: Telegram: @CBSETOPPERS

1

1 Unit I: Reproduction (Unit VI) 3
1.1 Sexual Reproduction in Flowering Plants . . . 3
1.1.1 Pollen-Pistil Interaction very high . . . 3
1.1.2 Double Fertilization very high . . . 3
1.1.3 Apomixis high . . . 4
1.2 Human Reproduction . . . 4
1.2.1 Gametogenesis very high . . . 5
1.2.2 Menstrual Cycle very high . . . 5
1.2.3 Fertilization and Implantation very high . . . 6
1.3 Reproductive Health . . . 6
1.3.1 Birth Control Methods very high . . . 6
1.3.2 MTP (Medical Termination of Pregnancy) high . . . 7
1.3.3 Assisted Reproductive Technologies (ART) very high . . . 7

2 Unit II: Genetics and Evolution (Unit VII) 8
2.1 Principles of Inheritance . . . 8
2.1.1 Mendel's Laws very high . . . 8
2.1.2 Incomplete Dominance & Codominance very high . . . 8
2.1.3 Chromosomal Disorders very high . . . 8
2.2 Molecular Basis of Inheritance . . . 9
2.2.1 DNA Structure — Watson-Crick Model very high . . . 9
2.2.2 Transcription very high . . . 9
2.2.3 Translation very high . . . 9
2.2.4 Lac Operon very high . . . 10
2.3 Evolution . . . 10
2.3.1 Homologous Organs & Adaptive Radiation high . . . 10
2.3.2 Hardy-Weinberg Principle high . . . 10

3 Unit III: Biology in Human Welfare (Unit VIII) 12
3.1 Human Health and Disease . . . 12
3.1.1 Life Cycle of Plasmodium high . . . 12
3.1.2 Immunity very high . . . 12
3.1.3 HIV-AIDS very high . . . 12
3.1.4 Cancer high . . . 13
3.2 Microbes in Human Welfare . . . 13
3.2.1 Sewage Treatment & BOD high . . . 13
3.2.2 Biogas Production medium . . . 13
3.2.3 Biocontrol Agents medium . . . 14

4 Unit IV: Biotechnology (Unit IX) 15
4.1 Biotechnology: Principles and Processes . . . 15
4.1.1 Restriction Enzymes very high . . . 15
4.1.2 Gel Electrophoresis very high . . . 15
4.1.3 PCR (Polymerase Chain Reaction) very high . . . 15
4.2 Biotechnology and Its Applications . . . 15

4.2.1 Bt-Cotton high . . . 16
4.2.2 RNA Interference (RNAi) high . . . 16
4.2.3 Gene Therapy high . . . 16

5 Unit V: Ecology and Environment (Unit X) 17
5.1 Populations: Growth Curves and Biotic Interactions . . . 17
5.1.1 Population Growth Curves high . . . 17
5.1.2 Biotic Interactions high . . . 17
5.2 Ecosystem: Productivity, Energy Flow, Nutrient Cycling . . . 17
5.2.1 Ecosystem Productivity & Energy Flow very high . . . 18
5.2.2 Nutrient Cycling high . . . 18
5.3 Biodiversity: The Evil Quartet and Conservation . . . 18
5.3.1 The Evil Quartet very high . . . 18
5.3.2 Conservation: In-situ and Ex-situ very high . . . 19

6 Quick Revision: Master Priority Summary 20

1

1

1.1.1 Pollen-Pistil Interaction very high

Definition

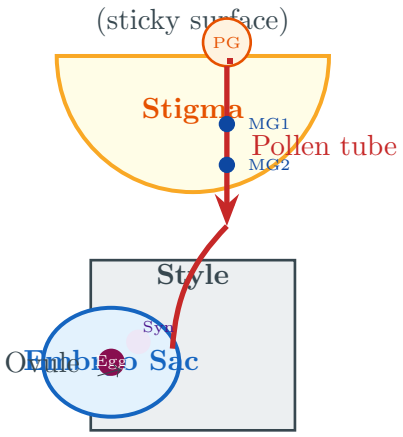
The ability of the **pistil** to recognize compatible pollen and initiate post-pollination events leading to fertilization.

Key Points:

- Pollen lands on stigma → rehydrated
- Stigma recognizes compatible/incompatible pollen
- Compatible pollen: pollen tube grows through **germ pores**
- Pollen tube enters ovule through **micropyle**
- Pollen tube carries 2 male gametes to embryo sac
- **Self-incompatibility**: pistil rejects own pollen (prevents inbreeding)

Outbreeding Devices:

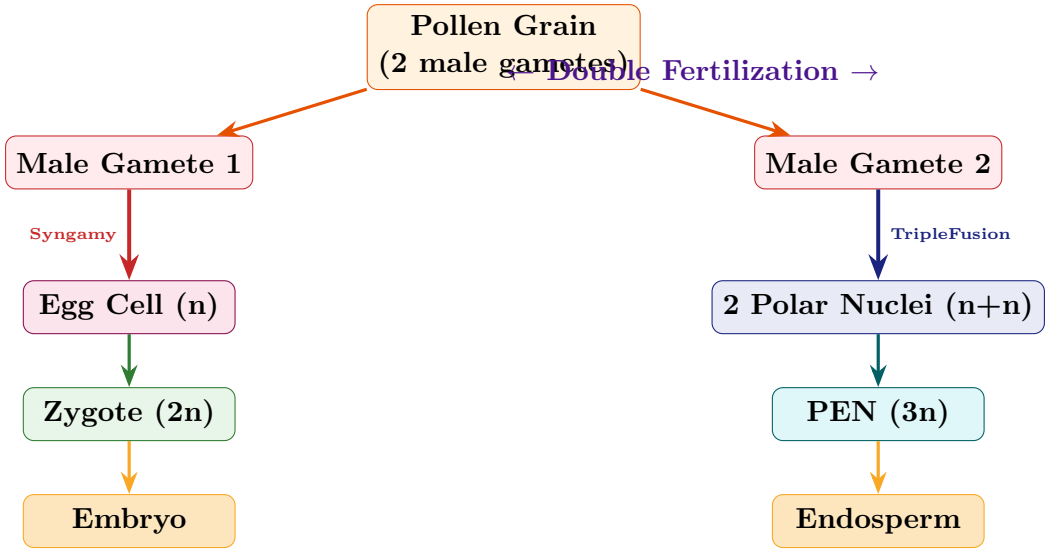
- Pollen release and stigma receptivity not synchronized
- Anther and stigma at different positions
- Production of unisexual flowers
- Self-incompatibility mechanisms



1.1.2 Double Fertilization very high

Definition

The process in which **two** types of fusion occur in the embryo sac — **Syngamy** (fertilization proper) and **Triple Fusion** — unique to angiosperms.



Key Points

- 1st MG + Egg → Syngamy → Zygote (2n)
- 2nd MG + 2 Polar nuclei → Triple Fusion → PEN (3n)
- PEN → Endosperm (food reserve)
- Zygote → Embryo
- Unique to angiosperms

Embryo Sac at Maturity (7-celled, 8-nucleate)

- 3 Antipodals (chalazal end)
- 2 Synergids + 1 Egg cell (micropylar = egg apparatus)
- 2 Polar nuclei in central cell

1.1.3 Apomixis high

Apomixis & Polyembryony

Apomixis:

- **Asexual reproduction** mimicking sexual reproduction
- Seeds formed **without fertilization**
- Diploid cell (no meiosis) → embryo
- Examples: **Citrus, Mango**
- Importance: Preserves hybrid characters

Polyembryony:

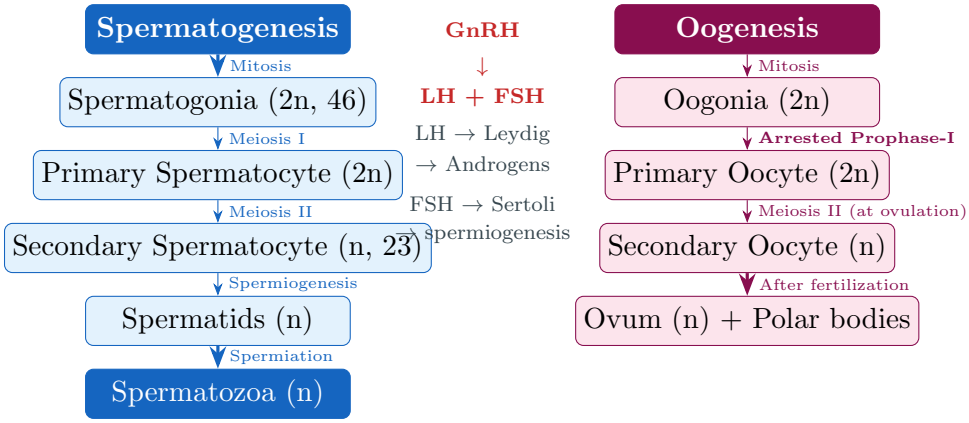
- More than one embryo per seed
- Often associated with apomixis
- Examples: **Citrus, Groundnut**
- Nucellus cells (2n) → protrude into embryo sac → embryos

Exam Tips — Ch.1

- **Draw Embryo Sac (labeled)** — always asked (3 marks)
- Double Fertilization flowchart — 2–3 marks guaranteed
- Microsporogenesis vs Megasporogenesis comparison table
- **Remember:** Sporopollenin (exine) = most resistant organic material

1

1.2.1 Gametogenesis very high



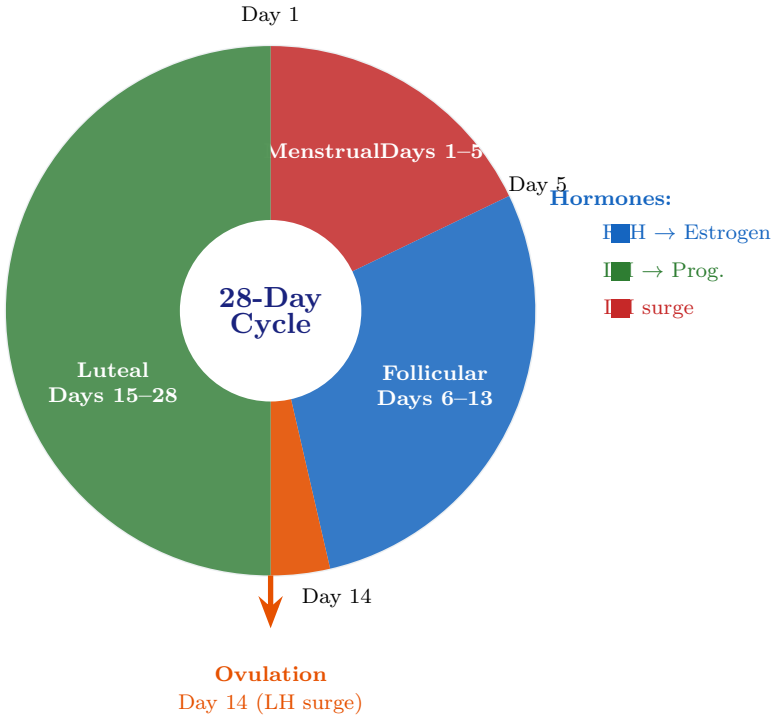
Sperm Structure

- **Head:** haploid nucleus + acrosome (hydrolytic enzymes)
- **Neck:** connects head to middle piece
- **Middle piece:** mitochondria (energy)
- **Tail:** flagellum for motility
- ~200–300 million sperms ejaculated per coitus

Follicular Development

- Oogonia → Primary follicle (granulosa cells)
- → Secondary follicle (more layers)
- → Tertiary follicle (antrum)
- → **Graafian follicle** → Ovulation
- Puberty: 60,000–80,000 primary follicles per ovary

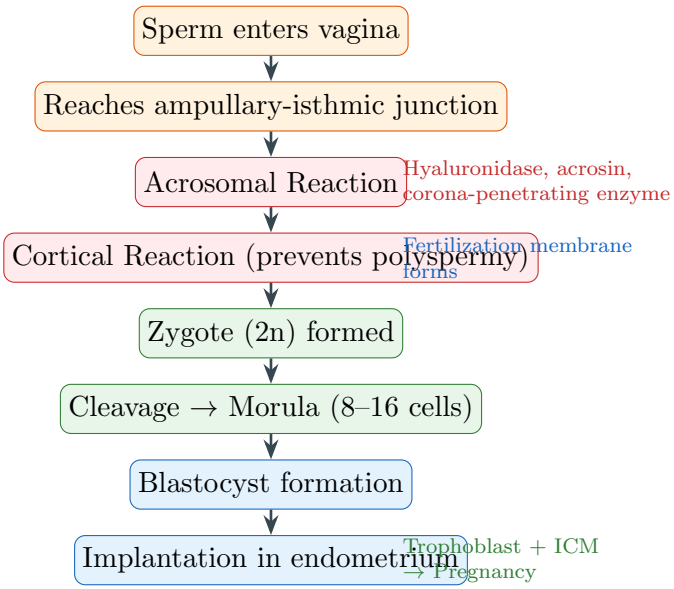
1.2.2 Menstrual Cycle very high



All 4 Phases of Menstrual Cycle

Phase	Days	Hormones	Events
Menstrual	1–5	LH falls, Prog.↓	Endometrium breaks down
Follicular	6–13	FSH → Estrogen	Endometrium proliferates
Ovulatory	14	LH surge	Graafian follicle ruptures
Luteal	15–28	Corpus luteum → Progesterone	Endometrium thickens

1.2.3 Fertilization and Implantation very high



Embryonic Development Timeline

Month 1 Heart forms
Month 2 Limbs and digits
3 months Major organs + external genitalia
5 months First foetal movement (quickening)
6 months Fine hair, eyelids form
9 months Fully developed foetus

1

1.3.1 Birth Control Methods very high

Contraceptive Methods — Complete Table

Category	Method	Key Detail
Natural	Periodic abstinence	Avoid coitus days 10–17
	Withdrawal	Male withdraws before ejaculation
	Lactational amenorrhea	No ovulation during intense breastfeeding
Barrier	Male condom	Latex; also prevents STDs
	Female condom	Diaphragm, cervical cap, vault
IUDs	Lippes loop	Non-medicated
	CuT, Cu7, MLS-375	Cu reduces sperm motility
	Progestasert, LNG-20	Hormone; prevents implantation
Oral pills	Combined pill	Progesterone + estrogen; inhibit ovulation
	Emergency	Within 72 hours of coitus
Surgical	Vasectomy	Vas deferens cut/tied (male)
	Tubectomy	Fallopian tube cut/tied (female)

1.3.2 MTP (Medical Termination of Pregnancy) high

MTP Key Facts

- Voluntary termination of pregnancy before full term = **induced abortion**
- **Legalized in India: 1971**
- Safe in **first trimester** (up to 12 weeks)
- Second trimester MTPs are much riskier

1.3.3 Assisted Reproductive Technologies (ART) very high

ART — Full Forms + Definitions (Memorize All)

Abbreviation	Full Form	Key Point
IVF	In Vitro Fertilization	Fertilization outside body; test tube baby
ZIFT	Zygote Intra Fallopian Transfer	Zygote / $\leq$8 blastomeres $\rightarrow$ fallopian tube
IUT	Intra Uterine Transfer	>8 blastomeres $\rightarrow$ uterus
GIFT	Gamete Intra Fallopian Transfer	Donor gametes $\rightarrow$ recipient's fallopian tube
ICSI	Intra Cytoplasmic Sperm Injection	Sperm directly injected into ovum
IUI	Intra Uterine Insemination	Donor semen $\rightarrow$ uterus

Exam Tips — Reproductive Health

ART full-forms table is asked **every year** (3 marks). Vasectomy vs Tubectomy = 1 mark. MTP year (1971) = 1 mark. Side effects of oral pills = 1 mark.

1

1

2.1.1 Mendel's Laws very high

Mendel's 3 Laws

1. **Law of Dominance:** In heterozygote, only dominant allele expresses.
2. **Law of Segregation (Purity of Gametes):** Alleles separate during gamete formation; each gamete gets only ONE allele.
3. **Law of Independent Assortment:** Alleles of different genes assort independently (proven by dihybrid cross).

Monohybrid Cross

	T (sperm)	t (sperm)
T (egg)	TT	Tt
t (egg)	Tt	tt

Phenotypic: **3 Tall : 1 Dwarf**
Genotypic: **1 TT : 2 Tt : 1 tt**

Dihybrid F_2 Ratio

9 R_Y	3 R_yy	
3 rrY_	1 rryy	

9:3:3:1

2.1.2 Incomplete Dominance & Codominance very high

Incomplete Dominance

- Neither allele fully dominant
- **Intermediate phenotype** in F1
- F2: 1 Red : 2 Pink : 1 White
- Genotypic = Phenotypic ratio (1:2:1)
- Example: **Mirabilis jalapa / Antirrhinum** (snapdragon)

Codominance & Multiple Alleles

- Both alleles express equally
- Example: **ABO Blood Groups**
- 3 alleles: I^A, I^B, i
- I^A and I^B are codominant
- Blood group AB: both express simultaneously

2.1.3 Chromosomal Disorders very high

Chromosomal Disorders — Comparison Table

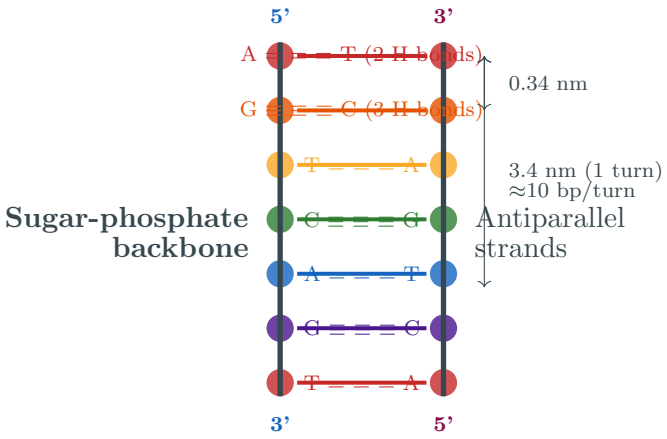
Disorder	Chromosomes	Karyotype	Features
Down's Syndrome	Trisomy 21	47	Short, small rounded head, mental retardation
Klinefelter's	Extra X	47 (XXY)	Male, gynaecomastia, sterile
Turner's Syndrome	Missing X	45 (XO)	Female, rudimentary ovaries, sterile

Mendelian Disorders:

- **Haemophilia:** X-linked recessive; non-stop bleeding; carrier female → hemophilic son
- **Sickle Cell Anaemia:** Autosomal recessive; GAG→GUG; Glu→Val at position 6 of β-globin; sickle-shaped RBCs
- **PKU:** Autosomal recessive; missing phenylalanine hydroxylase → mental retardation
- **Color Blindness:** X-linked recessive; cannot distinguish red-green

1

2.2.1 DNA Structure — Watson-Crick Model very high

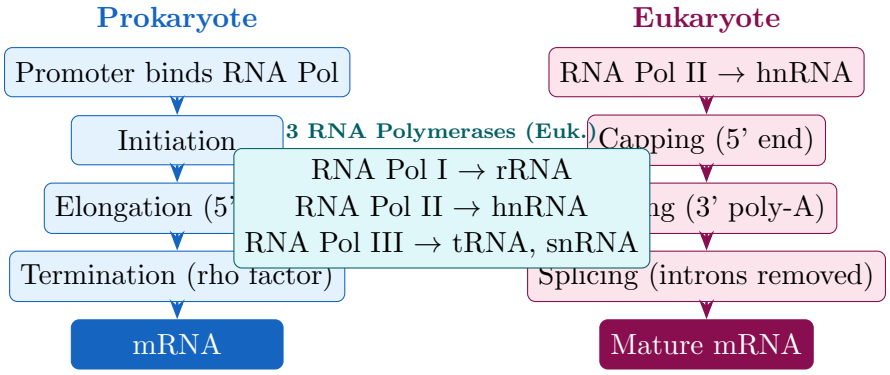


Watson-Crick DNA Double Helix

DNA Double Helix Key Features

- Two antiparallel polynucleotide chains
- Backbone: sugar-phosphate; bases project inward
- A=T (2 H-bonds); G≡C (3 H-bonds)
- **Chargaff's rule:** $[A] = [T]$; $[G] = [C]$
- Right-handed helix
- Pitch = 3.4 nm; ∼10 bp per turn
- Proposed by Watson & Crick (1953)
- X-ray data: Rosalind Franklin & Maurice Wilkins
- **Central Dogma:** DNA → RNA → Protein

2.2.2 Transcription very high



2.2.3 Translation very high

Translation — Key Points

tRNA (Adapter Molecule):

- Cloverleaf secondary structure
- Anticodon loop + amino acid acceptor arm
- Specific for each amino acid

Ribosome sites:

- A site (aminoacyl-tRNA)
- P site (peptidyl-tRNA)
- E site (exit)

Process:

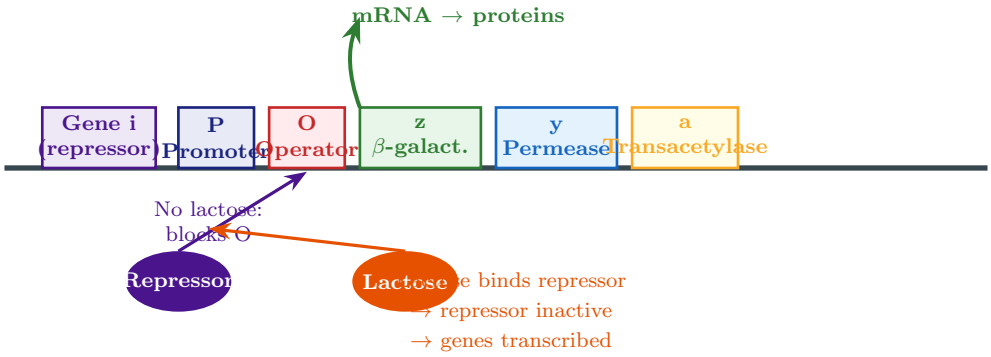
1. Charging of tRNA (aminoacyl-tRNA)
2. Initiation: ribosome binds AUG
3. Elongation: peptide bonds form
4. Termination: release factor at UAA/UAG/UGA

Genetic Code:

- Triplet, degenerate, non-overlapping
- Universal; AUG = start (Met)
- 61 sense + 3 stop codons

2.2.4 Lac Operon very high

Lac Operon — Negative Regulation (Inducible System)



1

2.3.1 Homologous Organs & Adaptive Radiation high

Homologous Organs

- Same structure, different functions
- **Divergent evolution**
- Examples: forelimbs of human, bat, whale, cheetah

Analogous Organs

- Different structure, same function
- **Convergent evolution**
- Examples: wings of bat and insect

Adaptive Radiation

Evolution of different species from a common ancestor to fill different ecological niches.

- **Darwin's finches** (Galapagos) — 14 species, different beaks
- **Australian marsupials** — all evolved from one ancestor on isolated continent

2.3.2 Hardy-Weinberg Principle high

Hardy-Weinberg Principle

Allele frequencies in a population remain constant generation to generation = genetic equilibrium.

Equation:

$$p + q = 1 \quad \text{(allele frequencies)}$$

$$p^2 + 2pq + q^2 = 1 \quad \text{(genotype frequencies)}$$

5 Factors that DISTURB equilibrium (= cause evolution):

1. Gene flow
2. Genetic drift
3. Mutation
4. Recombination
5. Natural selection

Founder effect: Small population colonizes new area → different allele frequencies.

Exam Tips — Genetics & Evolution

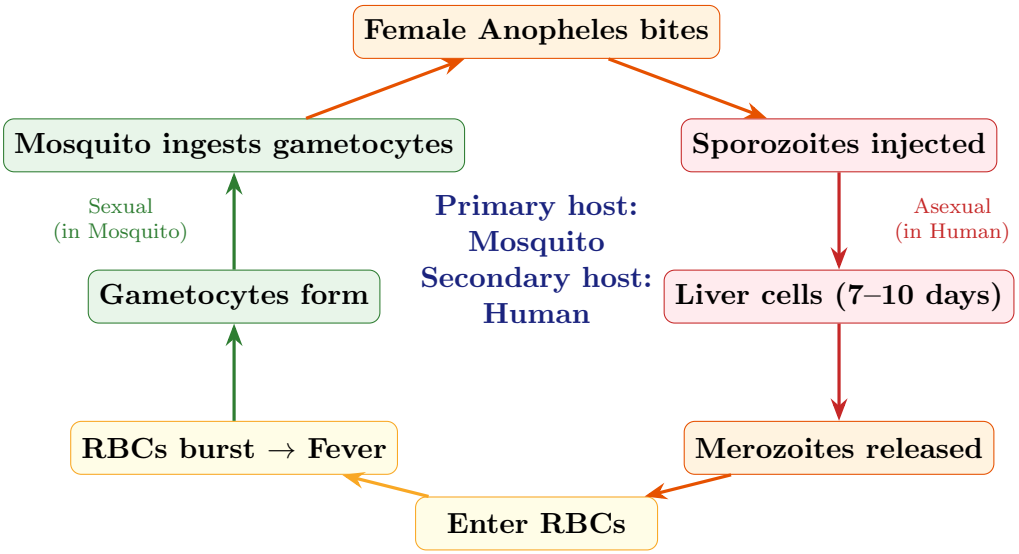
- Punnett squares (mono/dihybrid/sex-linked) = 3–5 marks GUARANTEED
- Chromosomal disorders comparison table = 3 marks
- Lac operon labeled diagram = 3–5 marks
- DNA replication diagram with all enzymes = 3–5 marks

- Hardy-Weinberg equation derivation = 2 marks

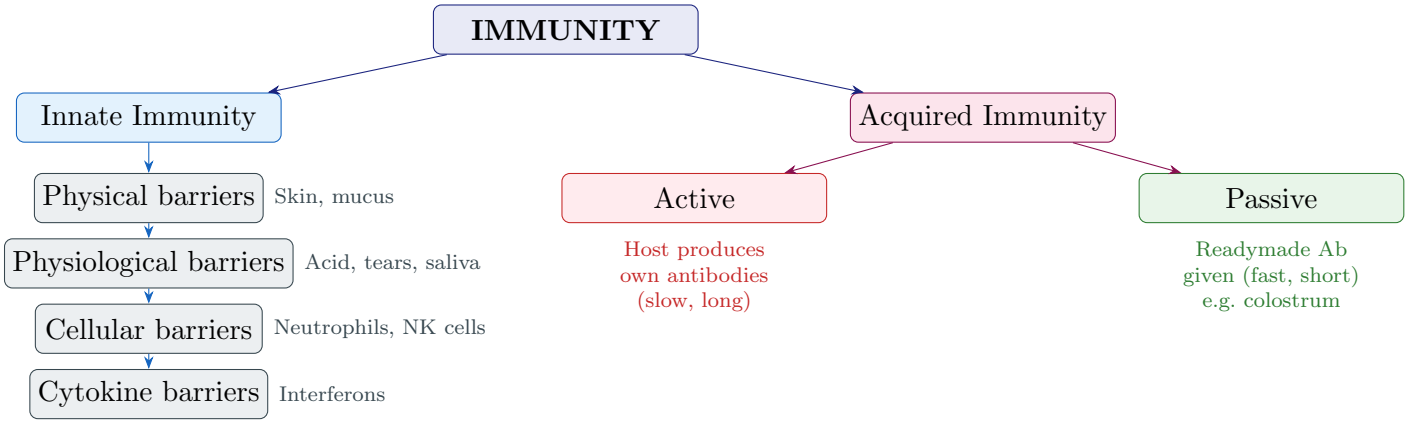
1

1

3.1.1 Life Cycle of Plasmodium high



3.1.2 Immunity very high



3.1.3 HIV-AIDS very high

HIV-AIDS — Complete Notes

Pathogen: HIV (retrovirus; genome = RNA)
First reported: 1981
Attacks: CD4+ Helper T-lymphocytes
Transmission:

- Sexual contact
- Contaminated blood/needles
- Mother to child (placenta)

Pathogenesis: HIV → macrophages → reverse transcriptase → viral DNA → host cell → more HIV → Helper T cells destroyed → immunodeficiency
Detection: ELISA → confirm by Western Blot
Treatment: Antiretroviral drugs (no cure)
Incubation: 5–10 years

3.1.4 Cancer high

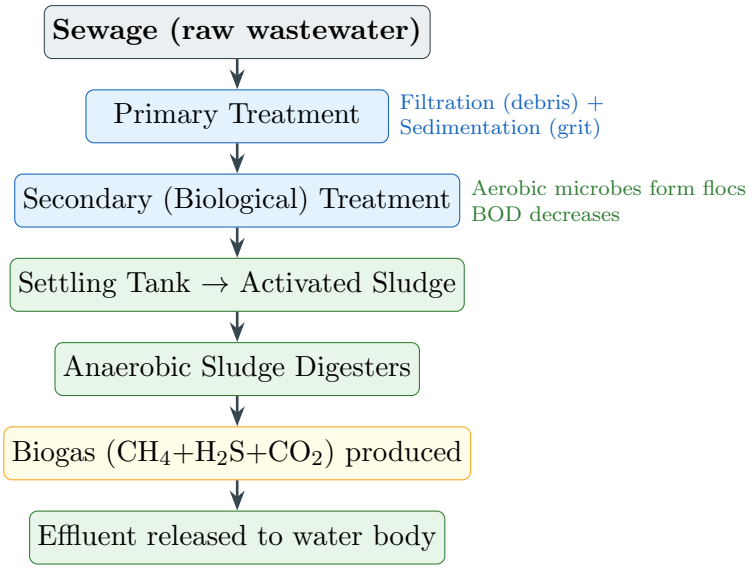
Cancer Key Points

Type	Benign	Malignant
Growth	Slow, limited	Fast, uncontrolled
Spread	Does not spread	Metastasis
Danger	Less	Life-threatening

Causes (Carcinogens): Ionizing radiation, UV, chemical carcinogens (tobacco), oncogenic viruses
Detection: Biopsy, CT, MRI, ELISA for cancer antigens
Treatment: Surgery + radiotherapy + chemotherapy + immunotherapy (α-interferon)

1

3.2.1 Sewage Treatment & BOD high



BOD (Biochemical Oxygen Demand)

Amount of O_2 consumed to oxidize all organic matter in 1 L water. **High BOD = more polluted water.**

3.2.2 Biogas Production medium

Biogas

- Produced by **methanogens** (e.g., *Methanobacterium*) from cellulosic material
- Composition: **CH_4 + CO_2 + H_2** (methane is main component)
- Cattle dung (gobar) rich in methanogens → **gobar gas**
- Technology by **IARI + KVIC**
- Spent slurry used as fertilizer

3.2.3 Biocontrol Agents medium

Biocontrol Agents

Organism	Type	Controls
Ladybird	Insect	Aphids
Dragonfly	Insect	Mosquitoes
Bacillus thuringiensis	Bacterium	Caterpillars on Brassica
Trichoderma	Fungus	Plant pathogens
Baculovirus (Nucleopolyhedrovirus)	Virus	Insects/arthropods (narrow-spectrum)

1

1

4.1.1 Restriction Enzymes very high

Restriction Endonucleases — Molecular Scissors

- Cut DNA at specific **palindromic sequences**
- Each enzyme recognizes a specific restriction site
- **Sticky ends:** complementary single-stranded overhangs (e.g., EcoRI)
- **Blunt ends:** no overhangs
- Examples: **EcoRI**, HindIII, BamHI
- Exonucleases: cut from ends
- Endonucleases: cut within DNA

EcoRI recognition site:

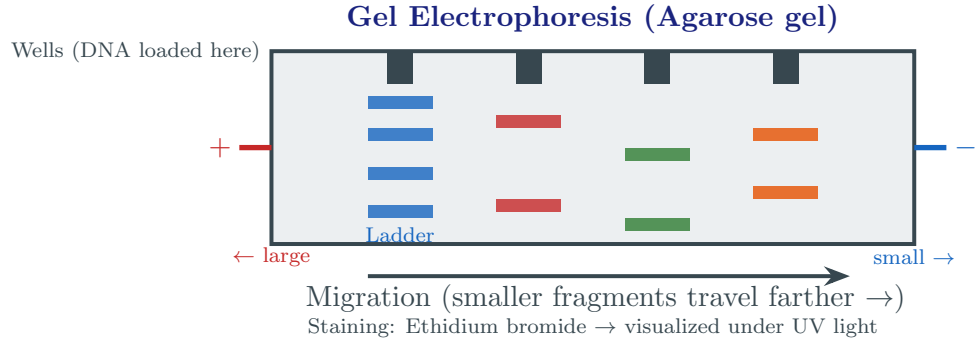
```
5'---G | AATTC---3'
3'---CTTAA | G---5'
```

After cutting:

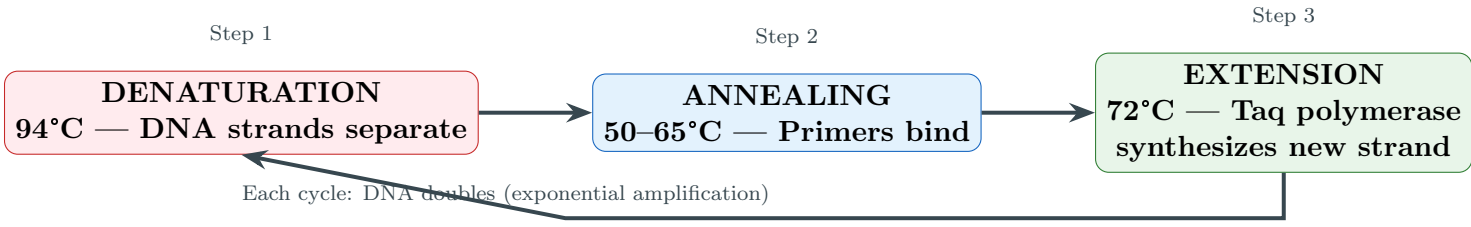
```
5'---G   AATTC---3'
3'---CTTAA   G---5'
```

(sticky ends formed)

4.1.2 Gel Electrophoresis very high



4.1.3 PCR (Polymerase Chain Reaction) very high



Needs: 2 primers (forward + reverse) + Taq polymerase (heat-stable)

1

4.2.1 Bt-Cotton high

Bt-Cotton — Mechanism

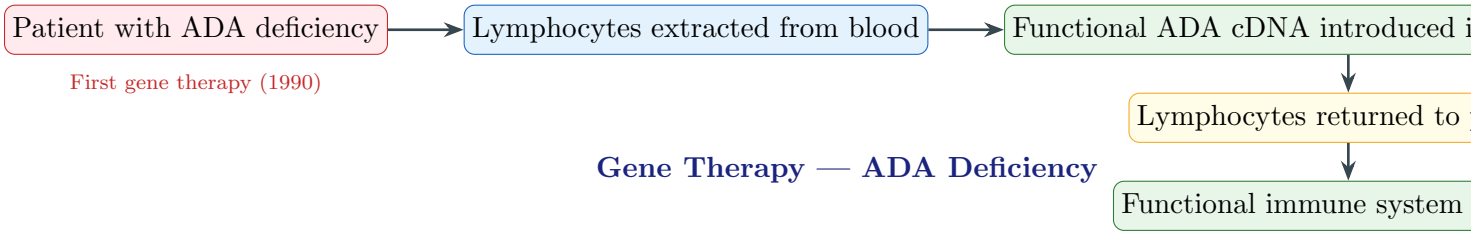
1. *Bacillus thuringiensis* produces **Cry proteins** (insecticidal) during specific growth phase
2. **cry** genes cloned and inserted into cotton genome
3. Insect ingests inactive protoxin → alkaline gut pH activates toxin
4. Activated toxin binds to midgut epithelial cells → pores form → cell swells and lyses → insect dies
5. **cryIAc, cryIIAb** → cotton bollworm; **cryIAb** → corn borer

4.2.2 RNA Interference (RNAi) high

RNA Interference

- Mechanism: double-stranded RNA (dsRNA) silences specific mRNA (prevents translation)
- Example: Used to protect tobacco from nematode *Meloidogyne incognita*
- nematode-specific dsRNA introduced into plant → nematode gene silenced → nematode dies
- Used in gene therapy and pest control

4.2.3 Gene Therapy high



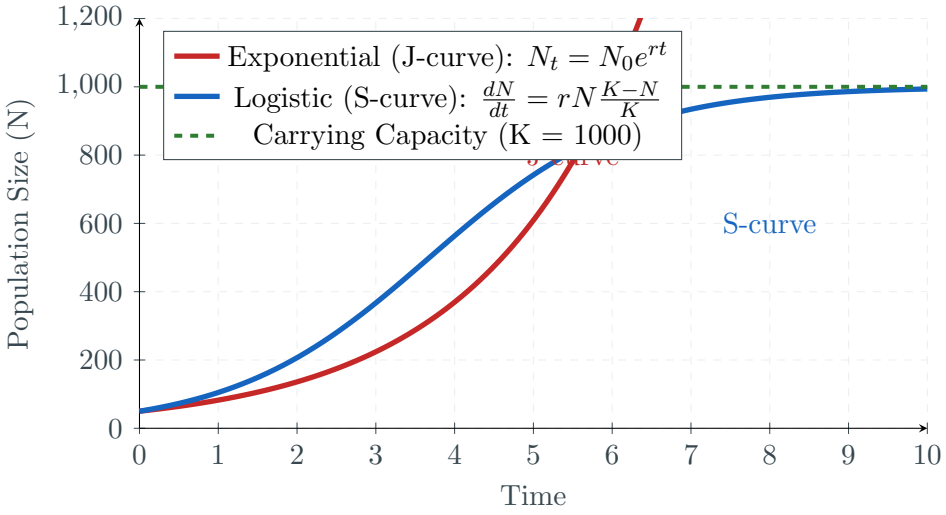
Exam Tips — Biotechnology

- Restriction enzymes (molecular scissors) + EcoRI palindrome = 1 mark
- PCR steps (3 steps, temperatures, enzyme) = 3 marks
- Gel electrophoresis labeled diagram = 3 marks
- Bt-cotton mechanism (cry genes, Cry proteins) = 3 marks
- Gene therapy (ADA, year 1990) = 2 marks

1

1

5.1.1 Population Growth Curves high



Growth Models Comparison

Feature	Exponential	Logistic
Resources	Unlimited	Limited
Shape	J-curve	S-curve (sigmoid)
Formula	$N_t = N_0 e^{rt}$	$\frac{dN}{dt} = rN\frac{(K-N)}{K}$
Observed in	Early colonization	Most natural populations
Carrying capacity	No K	K exists
Strategy	r-strategists	K-strategists

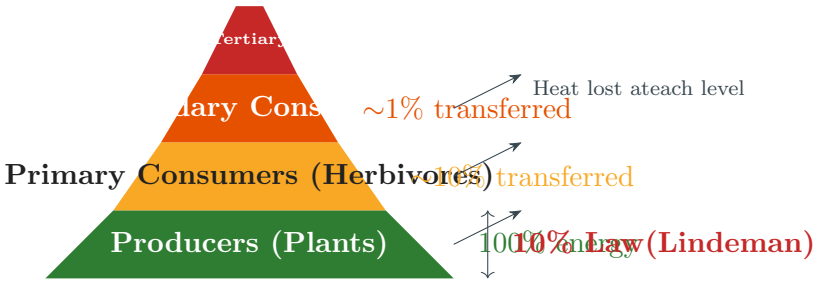
5.1.2 Biotic Interactions high

Population Interactions (+/− notation)

Interaction	Species A	Species B	Example
Mutualism	+	+	Mycorrhiza, Rhizobium-legume, lichen
Commensalism	+	0	Orchid on tree, clown fish in anemone
Predation	+	–	Lion-deer, Starfish-mussel
Parasitism	+	–	Plasmodium, tapeworm
Competition	–	–	Gause's competitive exclusion
Amensalism	0	–	*Penicillium* vs bacteria

1

5.2.1 Ecosystem Productivity & Energy Flow very high



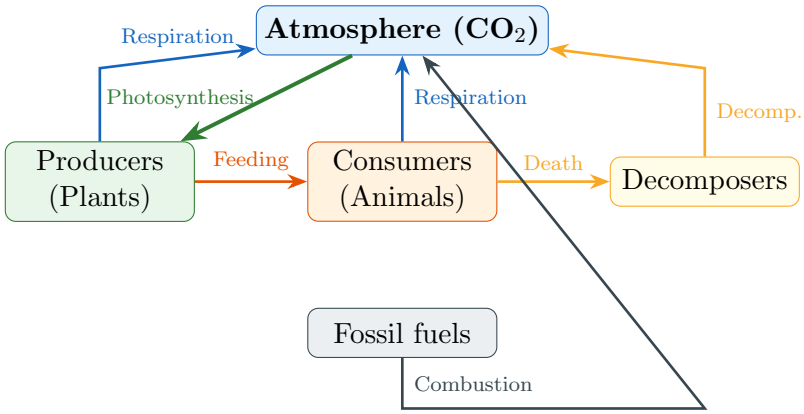
Ecological Pyramid & 10% Law of Energy Transfer

Productivity Types

Term	Definition
Gross Primary Productivity (GPP)	Total rate of photosynthesis including respiration losses
Net Primary Productivity (NPP)	GPP – Respiration = available for consumers
Secondary Productivity	Rate of biomass formation by consumers
Standing Crop	Total organic material (biomass) at a given time

5.2.2 Nutrient Cycling high

Carbon Cycle (simplified)



1

5.3.1 The Evil Quartet very high

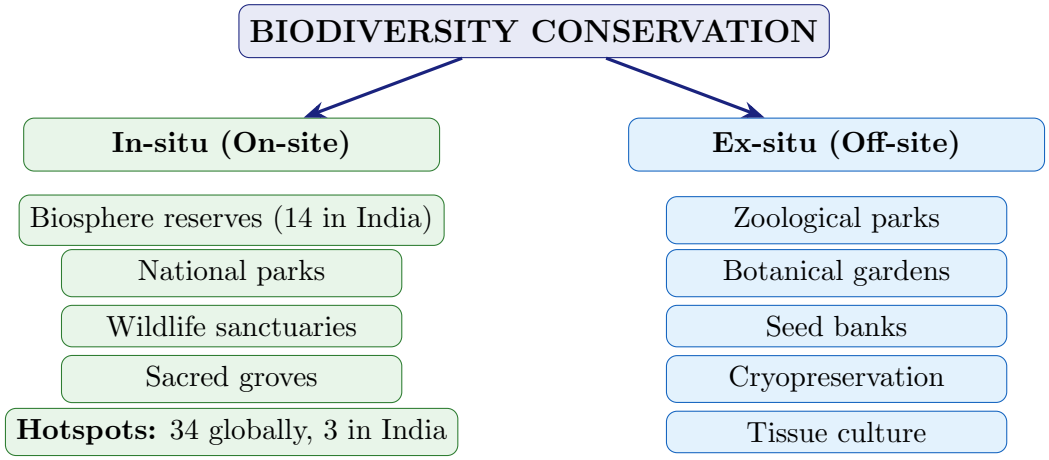
4 Main Causes of Biodiversity Loss — "The Evil Quartet"

1. **Habitat loss and fragmentation** — **MAJOR CAUSE**; deforestation, urbanization
2. **Overexploitation** — overfishing, hunting
3. **Alien / invasive species introduction** — water hyacinth, Nile perch, African catfish
4. **Co-extinctions** — mutualistic partner extinction leads to linked extinction

Types of Biodiversity:

- α (Alpha): within habitat
- β (Beta): between habitats
- γ (Gamma): landscape/regional

5.3.2 Conservation: In-situ and Ex-situ very high



Biodiversity Hotspots in India

1. **Western Ghats + Sri Lanka**
2. **Himalaya**
3. **Indo-Burma**

Criteria for Hotspot: >1500 endemic plant species AND >70% original habitat destroyed.

Exam Tips — Ecology

- Logistic vs exponential growth curve diagram = 3 marks (draw & label K)
- Biotic interactions table (6 types with +/−) = 3 marks
- Evil Quartet = 2 marks (name all 4)
- In-situ vs Ex-situ table = 3 marks
- Hotspots in India (3 names) = 1 mark
- Energy flow + 10% law (Lindeman) = 3 marks

1

Top 20 Must-Know Topics for Board 2025–26

Rank	Topic	Chapter	Marks	Priority
1	Double Fertilization + Embryo Sac	Ch.1	3–5	very high
2	Menstrual Cycle (all 4 phases)	Ch.2	3–5	very high
3	Spermatogenesis + Oogenesis	Ch.2	3–5	very high
4	Fertilization + Implantation	Ch.2	3–5	very high
5	ART — Full forms + Definitions	Ch.3	3	very high
6	Mendel's Laws + Crosses	Ch.4	5–8	very high
7	Chromosomal Disorders Table	Ch.4	3	very high
8	DNA Structure (Watson-Crick)	Ch.5	3–5	very high
9	DNA Replication (semi-conservative)	Ch.5	3–5	very high
10	Transcription (Prokaryote + Eukaryote)	Ch.5	3–5	very high
11	Translation + Genetic Code	Ch.5	3–4	very high
12	Lac Operon (diagram)	Ch.5	3–5	very high
13	HIV/AIDS pathogenesis + ELISA	Ch.7	3	very high
14	Immunity (Innate vs Acquired)	Ch.7	3	very high
15	Restriction Enzymes + Gel Electrophoresis	Ch.8	3–5	very high
16	PCR (3 steps, temperature)	Ch.8	3	very high
17	Bt-Cotton (Cry proteins, mechanism)	Ch.9	3	high
18	Biodiversity Conservation	Ch.13	3–5	very high
19	Ecological Pyramids + Energy Flow	Ch.12	3–5	high
20	Population Growth Models	Ch.11	3	high

Final Exam Strategy from TOPPERS 24/7

1. **Diagrams first:** Embryo sac, Double fertilization, Menstrual cycle, Spermatogenesis, Lac operon, DNA helix, PCR, Gel electrophoresis — each gives 1 extra mark
2. **1-mark questions:** Full forms (ART, PCR, ELISA, GMO, VNTR, HGP), years (MTP=1971, gene therapy=1990), names (Lindeman, Watson-Crick, Hershey-Chase)
3. **3-mark structure:** Definition + 2 key points + example
4. **5-mark structure:** Introduction + sub-headings + points + labeled diagram
5. **Revise NCERT:** These notes are supplementary — always refer to NCERT textbooks

Telegram: @CBSETOPPERS **Email:** toppers247@zohomail.in